

Criterion: Frontend Architecture & UX Delivery

Architecture Decision Record

Status

Status: Accepted

Date: 2026-01-05

Context

The project needs a responsive UI to browse/read disease content with search/filters and EN/RU support. The frontend should be fast to iterate and testable in a single-developer timeline.

Decision

Build a React + TypeScript SPA using Vite. Use React Router for navigation, Redux Toolkit for shared data flow, an Error Boundary + Sentry for crash handling, and SCSS modules for themeable styling.

Alternatives Considered

Alternative	Pros	Cons	Why Not Chosen
Next.js (SSR/ISR)	Better SEO defaults; SSR capabilities	More framework complexity; additional dependencies	Not a read-focused library where S...
No global store (local state only)	Less boilerplate for small apps	Harder cross-page caching and state management	Shared data (diseases/sources/risk fa...
UI framework (e.g., MUI)	Faster component assembly	Design constraints, heavier runtime	Project already uses SCSS modules and...

Consequences

Positive: - Fast dev experience and builds via Vite - Predictable state/data flow via Redux Toolkit - Centralized error capture (Sentry + Error Boundary) - Testability (Jest/RTL + Playwright; Storybook)

Negative: - SPA navigation requires explicit handling for SEO basics - Global store adds some complexity for small pages

Neutral: - Some routes/pages are scaffolded but not fully wired yet (see checklist)

Implementation Details

Project Structure

```
frontend/src/  
+-- app/  
|   +-- App.tsx  
|   +-- Router.tsx  
|   +-- providers
```

```

|   +-- layouts/MainLayout/           # Header + <Outlet /> shell
|   +-- providers/
|       +-- ErrorBoundary/           # React Error Boundary
|       +-- Router/                  # Route config + lazy pages
|       +-- StoreProvider/           # Redux store provider
+-- pages/                           # Route-level pages (Error/NotFound implemented)
+-- shared/
|   +-- api/                         # API client + slices/thunks
|   +-- context/                     # Theme context
|   +-- hooks/                       # useTheme and helpers
|   +-- ui/                          # Shared UI components (Header, Loader, etc.)
+-- assets/                         # SVG assets

```

Key Implementation Decisions

Decision	Rationale
BrowserRouter-based SPA	Simple routing for a content library
Lazy-loading for route pages	Reduce initial bundle; faster first p...
Global layout (MainLayout)	Consistent header + content; shared d...
Redux Toolkit store	Shared async data fetching and cachin...
Sentry + ErrorBoundary	Centralized error capture and user-fr...

Code Examples

Provider composition (routing + error boundary + store):

```

// frontend/src/app/App.tsx
<BrowserRouter>
  <ErrorBoundary>
    <StoreProvider>
      <Router />
    </StoreProvider>
  </ErrorBoundary>
</BrowserRouter>

```

Redux store assembly (RTK + RTK Query middleware):

```

// frontend/src/app/providers/StoreProvider/config/store.ts
const rootReducer = combineReducers({
  [api.reducerPath]: api.reducer,
  diseases: diseasesReducer,
  sources: sourcesReducer,
});

```

Diagrams

* High-level deployment/service interaction is documented in the technical deployment chapter.

Requirements Checklist

#	Requirement	Status	Evidence/Notes
1	SPA entypoint and routing found on	Open	frontend/src/app/App.tsx, frontend/sr...

2	Centralized layout shell	[OK]	frontend/src/app/layouts/MainLayout/M...
3	Shared store for API data	[OK]	frontend/src/app/providers/StoreProvi...
4	Error handling (user + monitoring)	[OK]	frontend/src/app/providers/ErrorBound...
5	Theme context available	[OK]	frontend/src/shared/hooks/useTheme.ts
6	Main library pages (Main/Source#1/Dise...		Routes exist but are currently commen...
7	Research cards/gallery route implemented		Route constant exists (/research), bu...
8	Automated testing for critical flows	[OK]	Jest + Playwright configured; tests p...

Known Limitations

Limitation	Impact	Potential Solution
Core content pages not mounted in router	Most user stories remain inaccessible...	Uncomment/implement MainPage, SourceP...
SPA SEO is limited by default	Discoverability may be reduced	Add sitemap/meta tags, and consider S...

References

- * frontend/package.json
- * frontend/src/app/App.tsx
- * frontend/src/app/providers/Router/ui/Router.tsx
- * frontend/src/app/providers/StoreProvider/config/store.ts
- * frontend/src/main.tsx